

Task 1.1: Automata Design

Regular Expressions for Each Token Category

- Integer Literal: $[+-]?[0-9]^+$
- Floating-point Literal: $[+-]?[0-9]^+\backslash\.[0-9]\{1,6\}([eE][+-]?[0-9]^+)?$
- Identifier: $[A-Z][a-z0-9_]\{0,30\}$
- Single-line Comment: $##[^\n]*$
- Boolean Literal: (true or false)
- Character Literal: $'([\^'\backslash\backslash n]\text{or}\backslash\backslash['\backslash\backslash ntr])'$
- Punctuator: $[(\{\}\backslash[\] , ; :)]$

NFA Diagrams

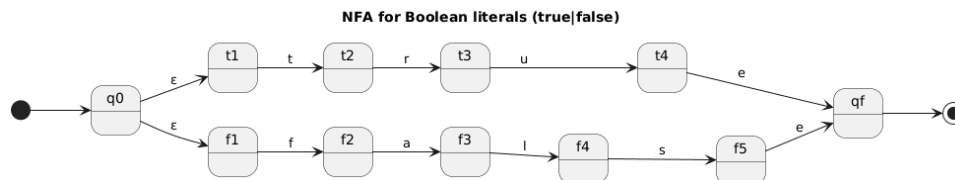


Figure 1: Boolean Literal NFA

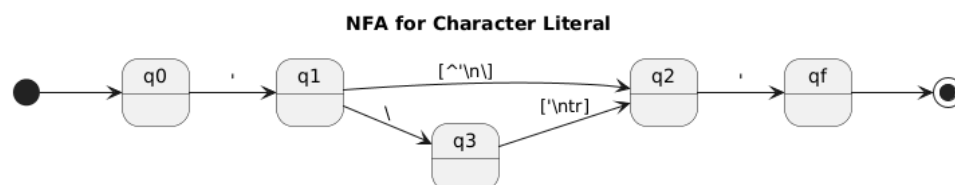


Figure 2: Character Literal NFA

NFA for single-line comment `##[^\n]*`

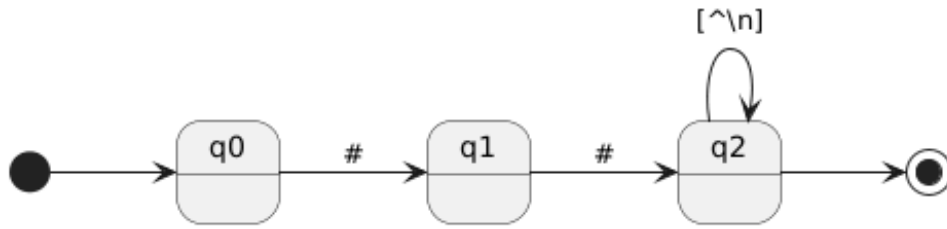


Figure 3: Single-line Comment NFA

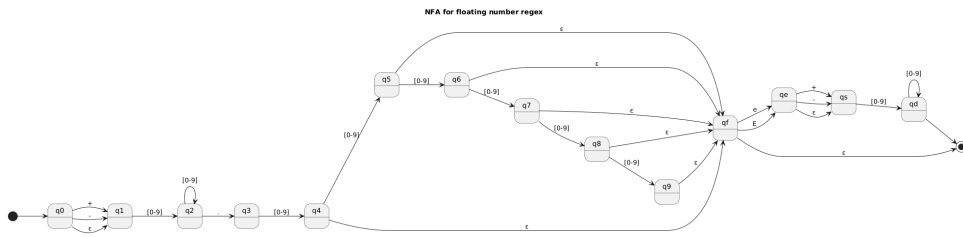


Figure 4: Floating-point Literal NFA

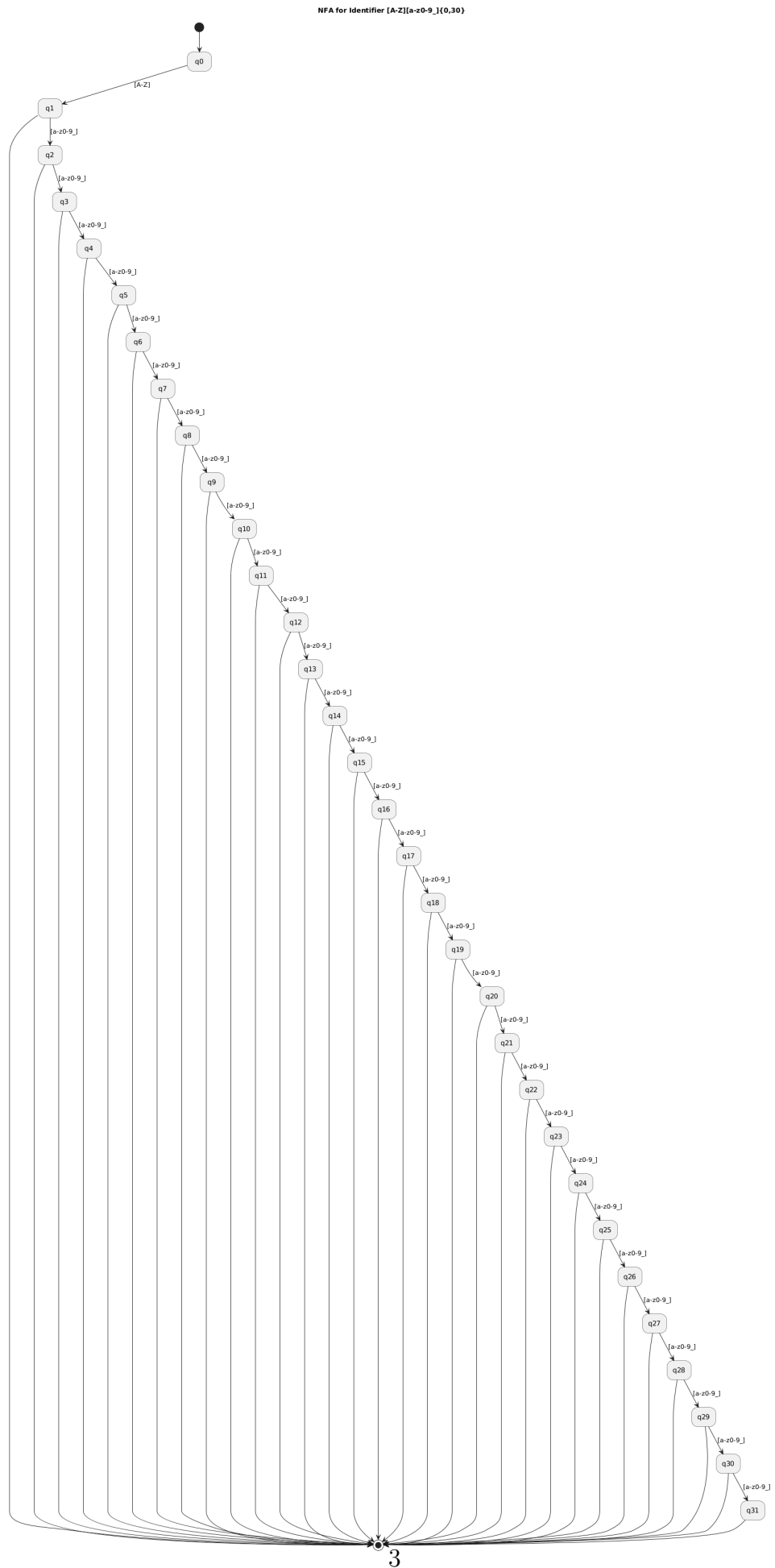


Figure 5: Identifier NFA

NFA for Punctuators

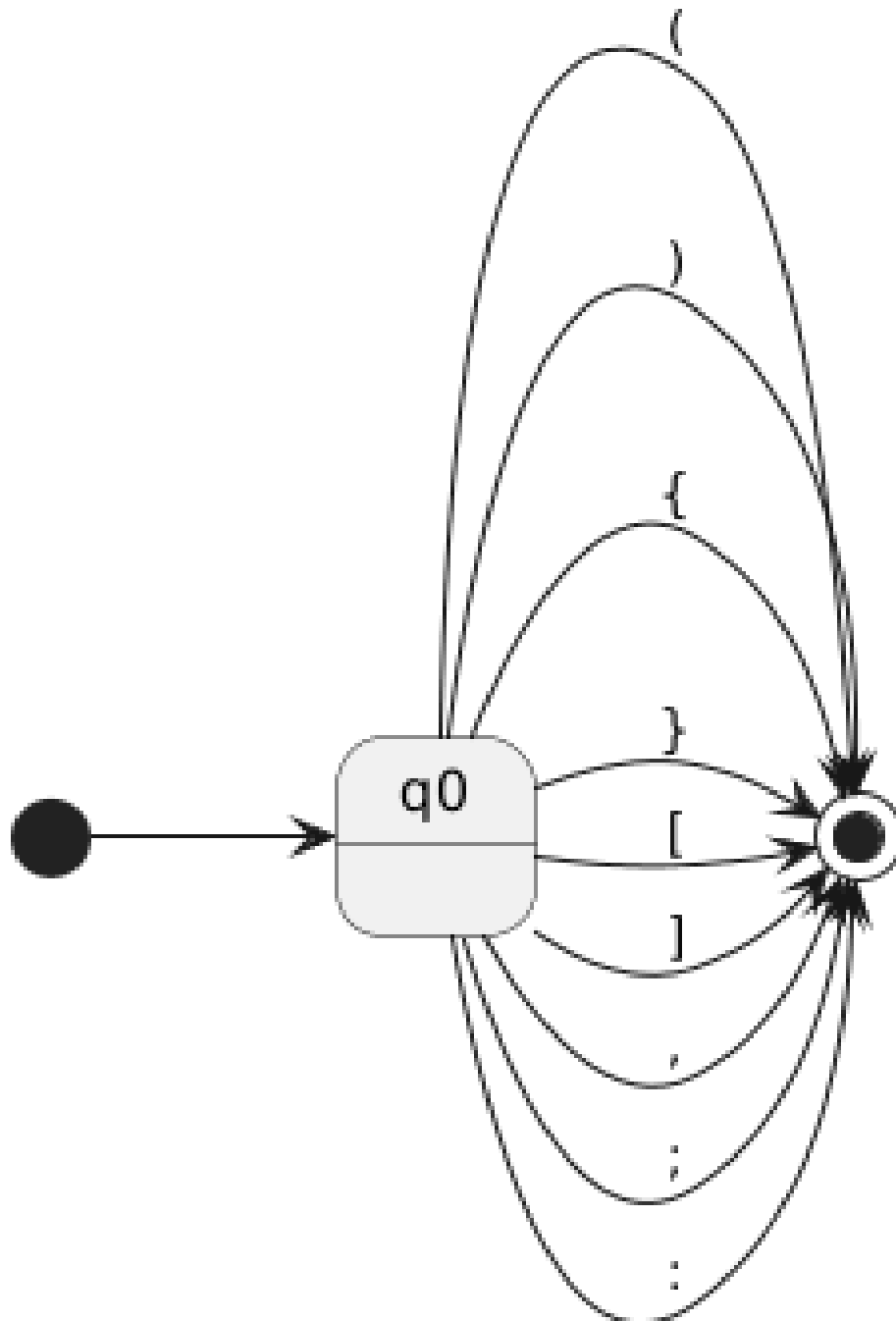


Figure 6: Punctuator NFA

NFA for regex: $[+-]?[0-9]^+$

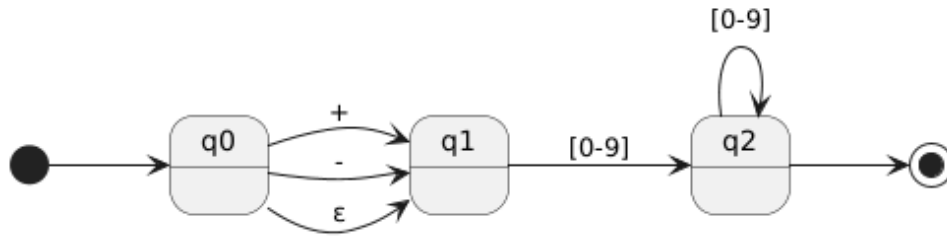


Figure 7: Integer Literal NFA

Minimized DFA Diagrams

Minimized DFA for Boolean literals (true|false)

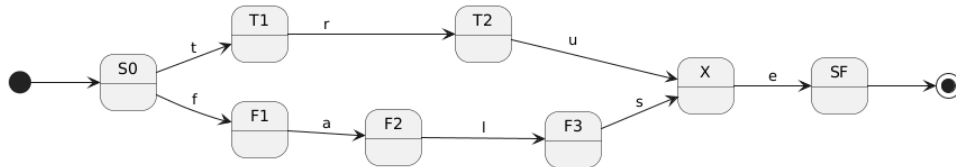


Figure 8: Boolean Literal DFA

DFA for Character Literal

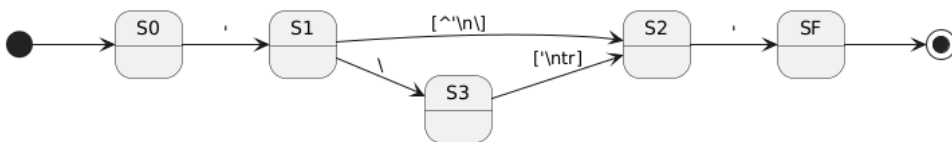


Figure 9: Character Literal DFA

DFA for single-line comment $##[^\\n]^*$



Figure 10: Single-line Comment DFA

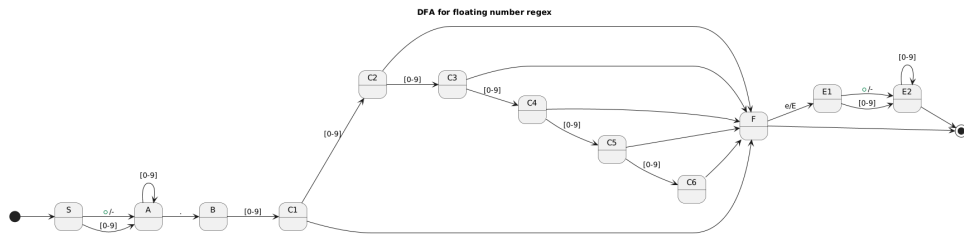


Figure 11: Floating-point Literal DFA

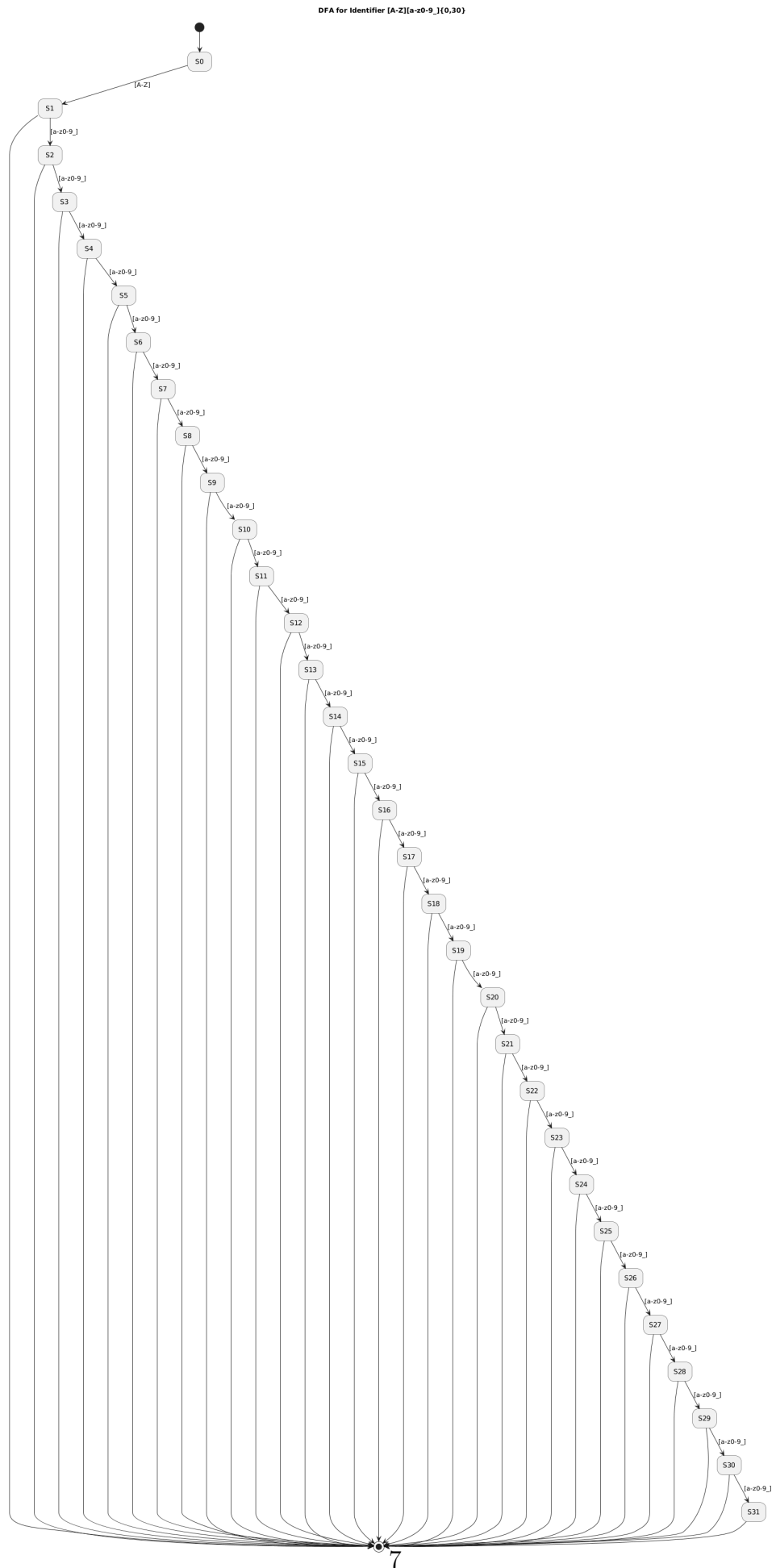


Figure 12: Identifier DFA

DFA for Punctuators

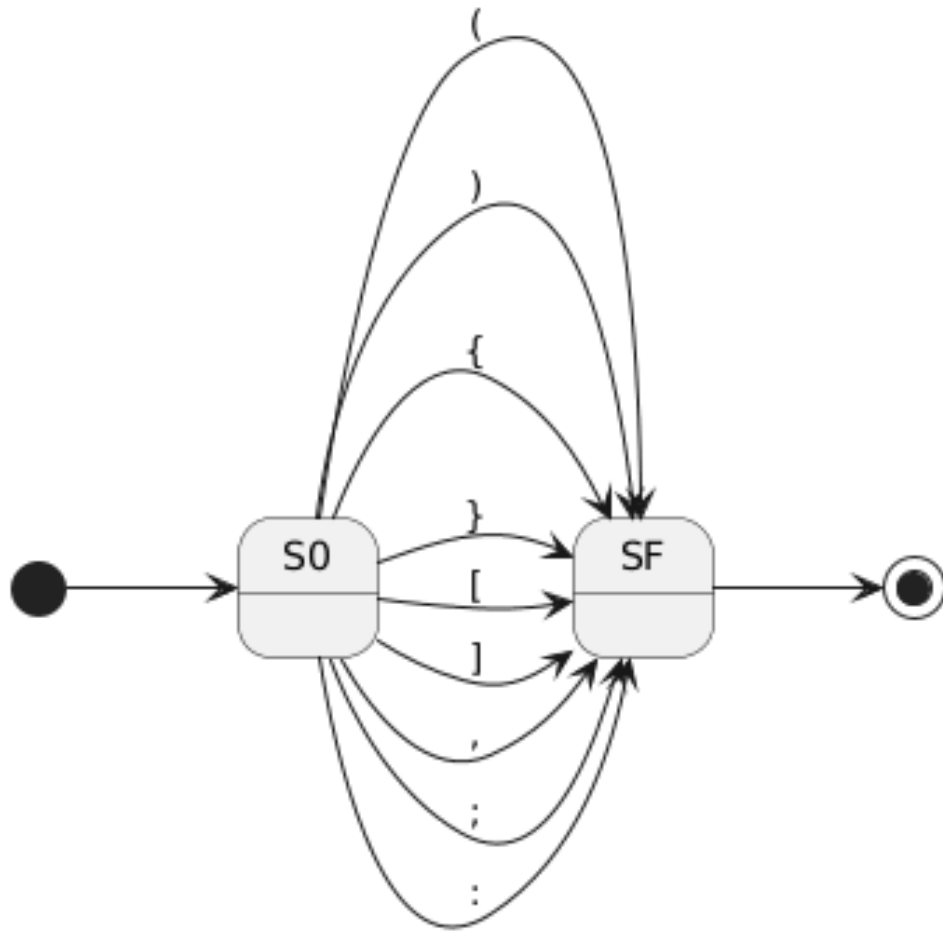


Figure 13: Punctuator DFA

DFA for regex $[+-]?[0-9]^+$

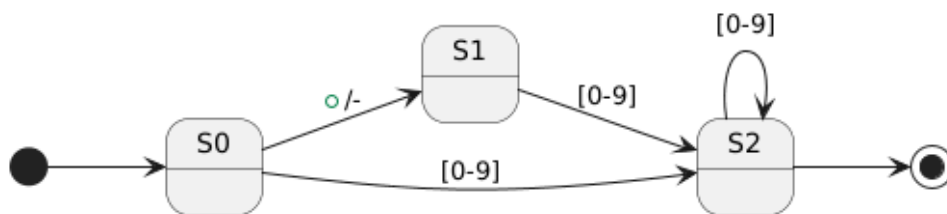


Figure 14: Integer Literal DFA